Masked Superpixel Contrastive Subspace Clustering Network for Unsupervised Large-Scale Hyperspectral Image Classification

Mengjiao Zhang, Tianhao Han, Xiaofeng Qu, Xizhan Gao, Xinwang Liu, *Senior Member, IEEE*, and Sijie Niu, *Member, IEEE*

***Abstract*—Subspace clustering (SSC) contributes a lot to the development of unsupervised hyperspectral image (HSI) classification task due to its ability of processing high-dimensional data. In general, SSC suffers from the bottlenecks of high computational cost when dealing with large-scale HSI. Some studies address this problem by using superpixel segmentation. However, the introduction of superpixel segmentation may cause some adverse effects, including interference from nontarget region samples and the oversmoothing of superpixel-level samples. To overcome these issues, we propose a masked superpixel contrastive subspace clustering (MSCSC) for large-scale HSI classification. Specifically, we leverage hyperspectral masked autoencoder (HMAE) rather than conventional autoencoder (CAE) as the backbone to mitigate the interference of nontarget region samples during feature extraction. Then, based on this backbone, we integrate contrastive learning (CL) with self-expressiveness-based SSC to form the superpixel-level contrastive SSC network, which aims to learn the discriminative superpixel-level self-representation. Moreover, we design a novel data augmentation strategy to ensure learn global and local information, improving the distinguishability of superpixel-level features. Experiments on four popular HSI datasets validate the superiority of our proposed method, with a large accuracy improvement compared with state-of-the-art methods.**

***Index Terms*—Contrastive learning (CL), deep subspace clustering (DSC), hyperspectral image (HSI), masked autoencoder (MAE), superpixel.**

I. Introduction

HYPERSPECTRAL image (HSI) are acquired through remote sensing techniques, which capture surface details at micrometer to nanometer spectral resolutions. HSI classification focuses on assigning each pixel to a specific land-cover categories and plays a crucial role in applications, including precision agriculture [1], [2], environmental monitoring [3], [4], and surface observation [5], [6]. Supervised methods [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17] have achieved remarkable results in HSI classification by leveraging labeled data to learn discriminative features. However, in practical applications, HSI is typically large scale, with larger sizes and higher pixel counts, making the annotation of extensive datasets a time-consuming and labor-intensive task. This limitation reduces the practicality of supervised methods in large-scale scenarios. In contrast, unsupervised methods, such as clustering [18], [19], [20], which can automatically classify similar HSI samples into the same cluster without manual labeling, are garnering increasing attention.

SSC is a popular clustering method which can group HSI samples from high-dimensional space to their latent low-dimensional subspaces. However, the HSI data are typically complex and do not subject to the linear assumption of SSC. Traditional SSC methods [21], [22], [23], [24], [25], [26] directly process the HSI data in original space, which can result in learning a low-quality self-expression matrix and restrict the effectiveness of clustering. Deep subspace clustering (DSC) methods [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37] make a progress by mapping the nonlinear HSI samples into linear feature spaces through convolutional neural network (CNN) and achieve considerable performance on relatively small datasets. Nevertheless, most SSC methods can hardly be applied to large-scale HSI because all the samples have to be fed into the models to learn self-expression coefficients and obtain the clustering results using singular value decomposition (SVD) [38], which leads to high time and computational consumption problems.

Recently, some SSC studies have been developed to solve this issue. There are mainly two ways to enhance the scalability: representative samples-based methods and superpixel segmentation-based methods. The core idea of the former is to select a part of representative samples from all samples to reduce the computational cost in self-expression learning. For instance, scalable sparse subspace clustering (SSSC) [39] selects a subset as in-sample data, and the rest of the samples are regarded as out-of-sample data to be represented as a combination of in-sample data to reduce the computational cost.

Received 2 January 2025; revised 22 April 2025, 10 June 2025, and 12 July 2025; accepted 17 July 2025. Date of publication 23 July 2025; date of current version 8 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62471202 and Grant 62302191; in part by the Natural Science Foundation of Shandong Province, China, through the Development Program Project of Youth Innovation Team of Institutions of Higher Learning in Shandong Province, under Grant ZR2023QF001; and in part by the Development Program Project of Youth Innovation Team of Institutions of Higher Learning in Shandong Province under Grant 2023KJ315. *(Corresponding authors: Xinwang Liu; Sijie Niu.)*

Mengjiao Zhang, Xiaofeng Qu, Xizhan Gao, and Sijie Niu are with the Shandong Key Laboratory of Ubiquitous Intelligent Computing and the School of Information Science and Engineering, University of Jinan, Jinan 250022, China (e-mail: zmj03222022@163.com; ise_quxf@163.com; gaoxizhan123@126.com; sjniu@hotmail.com).

Tianhao Han is with the School of AI and the Department of CSE, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: hth4511ujn@hotmail.com).

Xinwang Liu is with the School of Computer, National University of Defense Technology, Changsha 410073, China (e-mail: xinwangliu@nudt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3591899

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

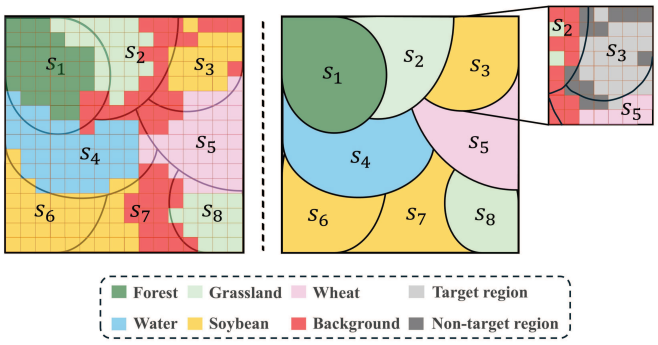


Fig. 1. Illustration of superpixel segmentation regions in HSI. (Left) Result of superpixel segmentation applied to the HSI, where the figure is divided into distinct regions (S_1–S_8). (Right) Assignment of a specific class label to each superpixel block. Specifically, the S_3 superpixel block is taken as an example. It is clearly observed that this region belongs to the soybean category, but simultaneously contains samples of background and wheat. In this superpixel block, the regions with soybean and wheat samples are regarded as target region, while the background sample area is regarded as nontarget region.

Similarly, selective sampling-based scalable sparse subspace clustering (S^5C) [40] also utilizes a subset to construct the affinity matrix and perform orthogonal iteration on eigenvalue decomposition. Spatial and cluster structural prior-guided subspace clustering (SPGSC) [41] selects landmark points and combines efficient regularizations to improve the clustering performance. However, due to the inherent uncertainty in the sample selection process, the chosen samples may fail to adequately represent the overall characteristics of the entire dataset, affecting the clustering performance. The latter way often achieves better performance, which utilizes a superpixel segmentation technique to aggregate the similar pixels in a contiguous area. Such as bipartite graph partition with graph nonnegative matrix factorization (BGP-GNMF) [42] and graph convolutional sparse SSC (GCSSC) [43] use superpixel segmentation as the anchor to alleviate the demand on memory space. By combining the excellent performance of DSC, superpixel contracted neighborhood contrastive subspace clustering (NCSC) [44] employs a superpixel pooling autoencoder to learn deep self-expression at the superpixel level and demonstrates state-of-the-art results.

Although the superpixel segmentation technique is widely used in large-scale HSI SSC, it also has some drawbacks. On the one hand, as shown in Fig. 1, superpixel blocks in HSI typically comprise target regions (consist of samples involved in classification evaluation) and nontarget regions (background and natural noise, collectively termed noise). Most deep learning methods construct a $w \times w$ window for each HSI pixel, called spatial-box, to better explore the spatial–spectral information. In target regions, spatial boxes are generally consistent and pure, enabling effective reconstruction of samples. However, in nontarget regions, they tend to be irregular and heterogeneous. The use of superpixels can unintentionally assign these nontarget regions to individual superpixel segments, degrading superpixel-level feature extraction. On the other hand, the original HSI is divided into numerous superpixel blocks through superpixel segmentation. These blocks are typically oversmoothing due to the use of an averaging strategy, which may result in the loss of detailed texture information.

To address the aforementioned challenges, we propose a novel masked superpixel contrastive subspace clustering (MSCSC) method for large-scale HSI clustering. Given the intrinsic spatial homogeneity of HSI, spatial-boxes extracted from target regions tend to exhibit uniform pixel distributions, whereas those from nontarget regions often present higher heterogeneity. To reduce the interference of nontarget regions on feature extraction, we design a lightweight masked autoencoder (MAE)-based backbone named hyperspectral MAEs (HMAEs). This design only excels at reconstructing homogeneous spatial-boxes from target regions but also effectively extracts useful information from highly complex nontarget region samples, boosting the model's capability to discriminate noise and capture intrinsic HSI features. Then, to obtain more distinguishable superpixel-level self-representations that are beneficial for downstream clustering task, we integrate contrastive learning (CL) with a self-expressiveness-based SSC framework, forming the superpixel contrastive subspace clustering (SCSC) network. The reason is that CL enhances feature separability by encouraging semantically similar samples to be closer and dissimilar samples to be farther apart in the latent space. The effectiveness of CL relies heavily on the construction of meaningful positive and negative sample pairs, and augmented samples should retain the spatial–spectral characteristics of the original HSI. Thus, we propose a novel data augmentation strategy, i.e., averaging and random selection, to enable the network capture both global and local contextual information from superpixel blocks.

1) We propose a MSCSC method for large-scale HSI classification, which enhances feature representations by effectively reducing the interference from nontarget region samples and alleviating the feature oversmoothing issues caused by superpixel segmentation.
2) We design a lightweight HMAE rather than simple conventional autoencoders (CAEs) as the backbone to enhance the capability of discriminating noise and capturing intrinsic HSI features, which effectively utilizes the intrinsic homogeneous properties of HSI.
3) We propose SCSC network to learn the discriminative superpixel-level self-representation. In this network, a data augmentation strategy of averaging and randomly selecting intrablock samples is proposed for CL to better capture the global and local information of HSI.
4) Experiments on four popular public HSI datasets demonstrate that our proposed approach surpasses most advanced large-scale HSI clustering algorithm with large accuracy improvement.

II. Related Works

A. Deep Learning for HSI Classification

HSI classification leverages the detailed spectral information from each band of HSI to accurately identify different types of land cover at the pixel level. Recently, deep learning methods have significantly advanced HSI classification by automatically learning hierarchical features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

CNN-based methods focus on local feature extraction through convolution operations, making them effective for capturing spatial features and local spectral information. Recently, many CNN-based methods have been proposed [44], [45], [46], [47]. For example, Li et al. [45] proposed a 3D-CNN-based approach for HSI classification. This method uses 3-D convolutions to capture joint spatial–spectral features, improving classification performance. However, 3D-CNNs come with high computational costs. As a result, hybrid models combining 2-D and 3-D convolutions have emerged. Roy et al. [46] proposed a hybrid spectral convolutional neural network (HybridSN) for HSI classification, which uses 2-D convolutions in the spatial dimension and 3-D convolutions in the spectral dimension, balancing computational efficiency and feature extraction capability. They perform well for processing high-resolution HSI data but struggle with modeling long-range dependencies and global spectral relationships.

Transformer-based methods have the ability to model long-range dependencies and can effectively capture the global features of the input data through the attention mechanism, which has attracted widespread attention from researchers. In recent years, an increasing number of methods have employed vision transformers (ViTs) [48] for supervised HSI classification, achieving remarkable performance. For example, Mei et al. [14] proposed a novel end-to-end group-aware hierarchical transformer (GAHT) and a new metric loss function CBML [15] to enhance the discriminative features of HSI samples. Xu et al. [16] proposed a multiscale and cross-level attention learning (MCAL) network and a cross-attention fusion (CAF) module [17] to fully explore global features of HSI.

B. Subspace Clustering

High-dimensional data are typically distributed as a union of multiple low-dimensional subspaces, posing significant challenges for subspace clustering. In the context of a dataset modeled as the union of k linear subspaces, subspace clustering aims to divide the data into distinct classes, with each class corresponding to an independent subspace. In general, subspace clustering methods comprise two steps: constructing an affinity matrix which captures the relationships between data, followed by the application of spectral clustering. Most research focuses on the first step to generate a high-quality affinity matrix, which influences clustering performance.

Elhamifar and Vidal [49] proposed one of the most classic subspace clustering (SSC) algorithms. Let $\mathbf{X} \in \mathbb{R}^{d\times n}$ be subject to a union of low-dimensional linear subspace, where n and d are the number and the dimensionality of data, respectively. $\mathbf{X} = \mathbf{XS} + \mathbf{E}$ represents the self-expression process, where $\mathbf{S} \in \mathbb{R}^{n\times n}$ is the self-expression coefficient matrix and $\mathbf{E}$ is the error matrix. The objective function of SSC is formulated as

$$\min_{\mathbf{S}} \|\mathbf{S}\|_0 \quad \text{s.t. } \mathbf{X} = \mathbf{XS} + \mathbf{E},\ \operatorname{diag}(\mathbf{S}) = 0 \tag{1}$$

where $\|\cdot\|_0$ refers to the l_0-norm, and the constraint condition $\operatorname{diag}(\mathbf{S}) = 0$ aims to prevent the occurrence of a trivial solution. Given the nonconvexity of the l_0-norm, the l_1-norm is often used as an approximation

$$\min_{\mathbf{S}} \|\mathbf{S}\|_1 \quad \text{s.t. } \mathbf{X} = \mathbf{XS} + \mathbf{E},\ \operatorname{diag}(\mathbf{S}) = 0. \tag{2}$$

Due to the powerful nonlinear feature extraction ability, DSC methods usually achieve better clustering performance than traditional one. As a typical DSC method, DSC-Net utilizes CAE to compute affinities among data points through a single fully connected (FC) self-expression layer. The objective function of DSC-Net is described as follows:

$$\min_{\mathbf{S}} \|\mathbf{S}\|_q + \frac{1}{2}\left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_F^2 \quad \text{s.t. } \operatorname{diag}(\mathbf{S}) = 0 \tag{3}$$

where $\|\cdot\|_q$ is a optional l_p-norm, $\hat{\mathbf{X}}$ is the reconstructed data through CAE and self-expression layer. Let $\boldsymbol{\Theta}$ be the parameters that need to be optimized for the entire network, the loss function used in DSC-Net is given as follows:

$$\mathcal{L}(\boldsymbol{\Theta}) = \frac{1}{2}\left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_F^2 + \lambda_1 \|\boldsymbol{\Theta}_S\|_q + \frac{\lambda_1}{2}\left\|\mathbf{Z}_{\boldsymbol{\Theta}_d} - \mathbf{Z}_{\boldsymbol{\Theta}_e}\boldsymbol{\Theta}_S\right\|_F^2 \tag{4}$$

where $\boldsymbol{\Theta}_S$ denotes the parameters of self-expression layer, $\boldsymbol{\Theta}_e$ denotes the parameters of encoder, $\boldsymbol{\Theta}_d$ denotes the parameters of decoder, $\mathbf{Z}_{\boldsymbol{\Theta}_e}$ denotes the low-dimensional features mapped by encoder, and λ_1 is a tradeoff parameter.

C. Masked Autoencoders

MAE is a form of autoencoder based on the ViT [48] architecture. By incorporating a masking mechanism, MAE masks a portion of the input data, compelling the model to learn feature representations from the remaining data, enhancing its ability to reconstruct the masked regions. Natural images often exhibit high spatial redundancy. For instance, without high-level understanding of image components, objects, or scenes, the network can effectively recover missing parts by leveraging information from adjacent image patches. The high masking ratio (e.g., 75%) and asymmetric encoder–decoder structure enable MAE to reduce computational costs significantly while achieving strong performance in top-layer unfrozen fine-tuning. Through pretraining with MAE, large data-demanding models can accelerate training and improve generalization performance.

Due to the superior performance on downstream tasks, some improved MAE-based methods have been utilized in various domains. Such as unsupervised anomaly detection [50], medical image analysis [51], and point clouds [52]. The flexibility and versatility of the MAE architecture make it as a valuable tool for HSI [53], [54].

D. Contrastive Learning

As deep learning and self-supervised learning advance, CL has become a key approach in representation learning. Many studies have made significant progress with CL. This method effectively learns HSI representations in unsupervised settings by constructing positive and negative sample pairs. Selecting an appropriate data augmentation strategy and constructing reasonable sample pairs are crucial for extracting useful representations for downstream tasks.

Several loss functions have been proposed for CL using positive and negative data pairs, including NCE loss [55], triplet loss [56], and NT-Xent loss [57]. For HSI clustering, the spectral–spatial contrastive clustering (SSCC) method

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

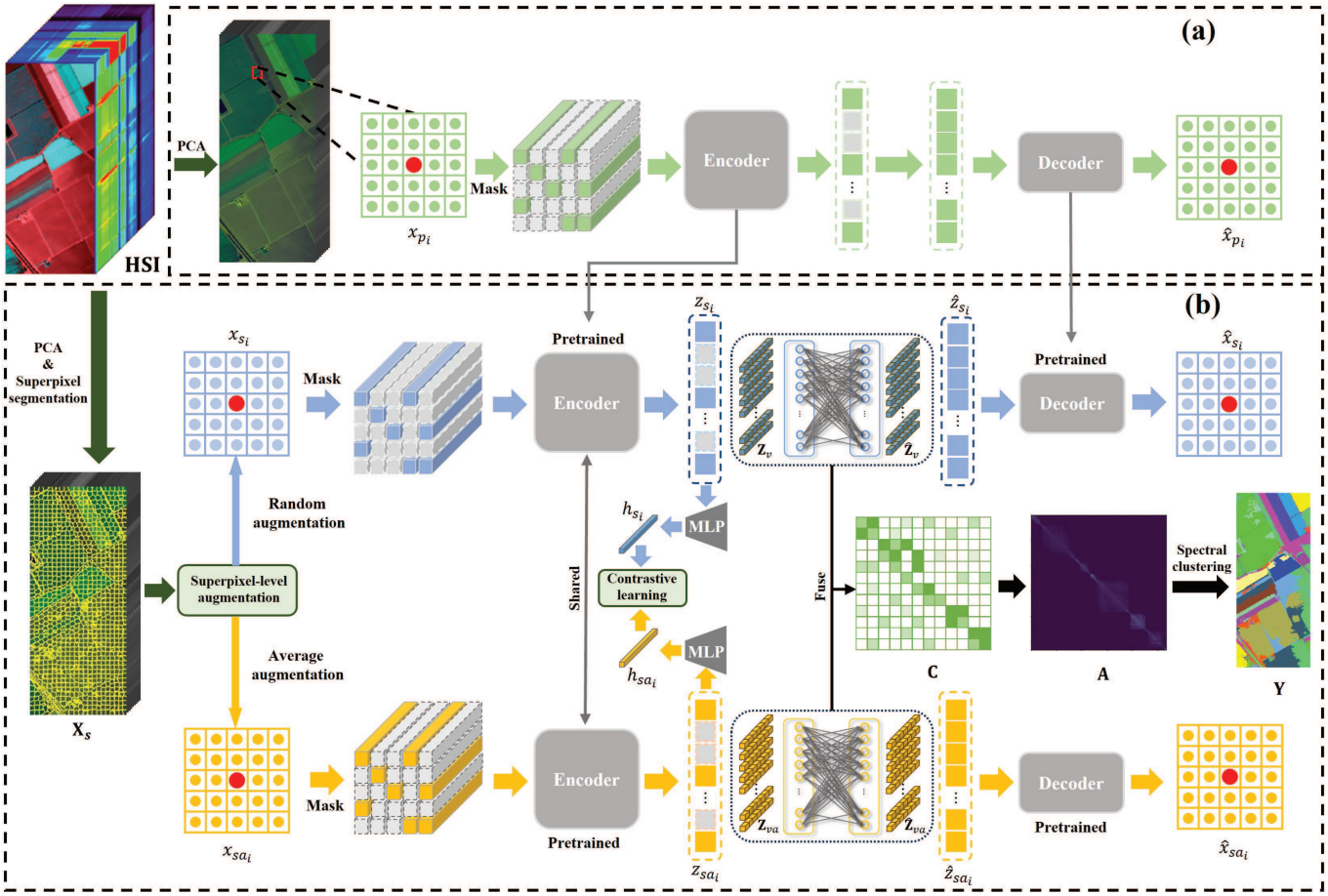


Fig. 2. Framework of the proposed MSCSC, including: (a) HMAE. The HMAE utilizes pixel-level spatial boxes processed by PCA as input to learn fine-grained HSI features and accurately capture intradomain characteristics, providing pretrained encoder and decoder for subsequent tasks. (b) SCSC network. This network shares parameters of ViT encoder and decoder from HMAE and learns more distinguishable superpixel-level self-representations. Then, we fuse the adaptive weights into a superpixel-level self-representation matrix, and get the final clustering results by spectral clustering.

[58] uses a spectral–spatial augmentation pool for dual CL. However, this method struggles to explore superpixel-level relationships between different homogeneous regions.

III. Proposed Method

In this section, we provide a detailed explanation of MSCSC. As shown in Fig. 2, MSCSC consists of two parts: HMAEs and SCSC network. First, HMAE is designed to replace the CAE as DSC backbone. Due to the intrinsic homogeneous properties of HSI, spatial patches in target regions are more consistent, while nontarget regions exhibit variability. The HMAE backbone can distinguish nontarget region samples and capture more intrinsic features of HSI. Second, SCSC network is formed by integrating CL with self-expressiveness-based subspace clustering, aiming to alleviate the oversmoothing and learn more discriminative superpixel-level self-representations. Furthermore, we design a novel data augmentation strategy that averages and randomly selects samples from one superpixel block to enhance the global and local information of HSI.

A. Preliminary

The DSC is an effective approach for classifying HSI. However, constructing self-representation matrix can generate excessive computational complexity, making it difficult to handle large-scale data in practical scenarios. At present, numerous DSC-based algorithms that use superpixel segmentation technique to solve the above challenge have been emerged. Despite making progress, we observe two issues.

1) Superpixel blocks usually contain both target and nontarget regions, inevitably containing samples from nontarget regions. Due to commonly used CAE only using 2-D convolutional receptive fields to capture local contextual information, they lack the ability to model long-range dependencies and dynamic weight adjustment mechanisms. This makes them ineffective at handling nontarget region interference, posing significant challenges for superpixel-level classification tasks.
2) Existing superpixel-based methods typically compute superpixel-level features by averaging the features within the superpixel block. This approach leads to the loss of detailed texture information, further limiting the discriminability of self-representation.

Given an HSI dataset $\mathbf{X} \in \mathbb{R}^{N\times D}$, where N denotes the number of pixels and D denotes the dimension. We preprocess $\mathbf{X}$ by: 1) reducing the dimensionality of D to D_p through principal component analysis (PCA) and 2) constructing the spatial-boxes $\mathbf{X}_p = [x_{p_1}, x_{p_2}, \ldots, x_{p_N}] \in \mathbb{R}^{N\times w\times w\times D_p}$ with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

spatial–spectral information by using a rolling window technique, where w is the window size of spatial-box.

B. Hyperspectral Masked Autoencoder

To capture the intrinsic features from superpixel blocks, we propose a lightweight MAE-based backbone named HMAE. The introduction of mask forces the model to rely on global context to infer the information of masked regions, which allows the model to comprehensively capture the features of the target regions in spatial patches while avoiding being easily influenced by local nontarget regions. The self-attention mechanism also learns the relationships between different positions within the same patch through long-range dependencies within spatial boxes, enabling the ability to identify important features in nontarget regions.

HMAE takes spatial-box x_p as the input, with a portion of its batches randomly masked. The masked spatial-box is fed into the ViT encoder to extract the feature representation, which is then sent into the ViT decoder to reconstruct the original spatial-box. Similar to other DSC methods, pretraining the HMAE can avoid directly training the whole network from scratch and accelerate the convergence. More importantly, it enhances the encoder's capability to learn fine-grained features of HSI and more accurately capture intradomain characteristics.

We denote the encoder parameters of HMAE as θ_E, decoder parameters as θ_D, and $f_{\theta_E}(\cdot)$ and $f_{\theta_D}(\cdot)$ are the operation of encoder and decoder, respectively. The pretraining process can be described as follows:

$$\hat{\mathbf{X}}_p = f_{\theta_D}\left(f_{\theta_E}\left(\mathbf{X}_p\right)\right) \tag{5}$$

where $\hat{\mathbf{X}}_p$ is the restored spatial-boxes.

The pretraining loss of HMAE can be formulated as follows:

$$\mathcal{L}_{\text{pretrain}} = \frac{1}{n_p}\left\|\mathbf{X}_p - \hat{\mathbf{X}}_p\right\|^2 \tag{6}$$

where n_p is the number of masked tokens.

C. Superpixel-Level Contrastive Subspace Clustering

To fully leverage the advantages of HMAE, a network based on a Siamese dual-branch architecture, named SCSC, is proposed. This network shares ViT encoder parameters and processes two distinct augmented views of superpixels through parallel branches, enabling it to learn discriminative self-representations. Specifically, the design enhances feature separability by utilizing intrasuperpixel consistency (positive pairs; samples from the same superpixel) and intersuperpixel differences (negative pairs; samples from different superpixels), learning more discriminative self-representations that improve the performance of the HSI classification task.

The superpixel-level feature extracted by ViT encoder needs to be fattened into a vector to perform self-expression learning and CL. Specifically, two FC layers are added between encoder and decoder to learn the superpixel-level self-expression property. To extract more informative superpixel-level features, SCSC reduces the distance between samples within the same superpixel block and increases the distance between samples from different superpixel blocks by CL. Furthermore, to obtain detailed texture information from HSI and further enhance the distinguishability of self-representation, we design a novel superpixel-level data augmentation strategy for the SCSC network.

For the superpixel processing, we employ simple linear iterative clustering (SLIC) [59] to generate the superpixel segmentation S for a large-scale HSI dataset. Supposing that N pixels are divided into M superpixel blocks, the superpixel region S can be defined as follows:

$$S = \{S_1, S_2, \ldots, S_M\}. \tag{7}$$

For a superpixel block S_i, we first select a pixel randomly and get its related spatial-box x_{s_i}, which is named random augmentation. Then, we compute the average of all the spatial-boxes x_{sa_i} whose corresponding pixels are in S_i. The average augmentation process can be formulated as follows:

$$x_{sa_i} = \frac{1}{|S_i|}\sum_{j=1}^{n_i} x_{S_i^j},\ s.t.\quad \{|S_i| = n_i\}_{i=1}^{M} \tag{8}$$

where $x_{S_i^j}$ denotes the spatial-box corresponding to the jth pixel in superpixel block S_i, n_i refers to the number of pixels within the ith superpixel block, and $|S_i| = n_i$ is the pixel count from S_i.

For each superpixel block S_i, we generate two augmented views. *Random Augmentation:* Randomly select a pixel within S_i and extract its spatial-box x_{s_i}. *Average Augmentation:* Compute the mean spatial-box $x_{sa_i} = (1/|S_i|)\sum_{j=1}^{n_i} x_{S_i^j}$ aggregated from all pixels in S_i. There are totally M^2 pair of spatial-blocks as the input of SCSC network, with M positive pairs (x_{s_i}, x_{sa_i}) from the same superpixel S_i and $M(M-1)$ negative pairs (x_{s_i}, x_{sa_j}) where $j \neq i$.

We transfer the parameters θ_E and θ_D of encoder and decoder from pretrained HMAE to initialize SCSC network. Subsequently, the spatial-box pairs are sent into the network to get the latent token features by $z_s = f_{\theta_E}(x_s)$ and $z_{sa} = f_{\theta_E}(x_{sa})$. Noted that the latent features z_s and z_{sa} consist of numerous vectors, we need to reshape z_s and z_{sa} as two single vectors $\mathbf{Z}_v$ and $\mathbf{Z}_{va}$. The following self-expression layer and decoder layer are used to obtain the self-expression matrix and reconstruct the original spatial-box, respectively. We can obtain the self-expression features by

$$\hat{\mathbf{Z}}_v = \mathbf{Z}_v\mathbf{C}_s \tag{9}$$

$$\hat{\mathbf{Z}}_{sa} = \mathbf{Z}_{va}\mathbf{C}_{sa} \tag{10}$$

where $\mathbf{C}_s$ and $\mathbf{C}_{sa}$ are the self-expression matrix of random augmented samples and average augmented samples, respectively. To reconstruct the input spatial-boxes, $\hat{\mathbf{Z}}_s$ and $\hat{\mathbf{Z}}_{sa}$ are obtained by separating into token features $\hat{\mathbf{Z}}_v$ and $\hat{\mathbf{Z}}_{va}$, respectively. The reconstructed spatial-boxes $\hat{\mathbf{X}}_s$ and $\hat{\mathbf{X}}_{sa}$ can be obtained by the decoder that $\hat{\mathbf{X}}_s = f_{\theta_D}(\hat{\mathbf{Z}}_s)$ and $\hat{\mathbf{X}}_{sa} = f_{\theta_D}(\hat{\mathbf{Z}}_{sa})$.

To optimize the similarities between each pair of samples without compromising generality, we stack a two-layer nonlinear MLP as the $g_C(\cdot)$ to map pairs of latent feature matrices to perform CL that $h_{s_i} = g_C(z_{s_i})$ and $h_{sa_i} = g_C(z_{sa_i})$.

When the training of SCSC network is completed, we can obtain two self-expression matrices $\mathbf{C}_s$ and $\mathbf{C}_{sa}$. Because

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

$\mathbf{C}_s$ and $\mathbf{C}_{sa}$ are from a different self-expression layer, we design a tradeoff parameter λ to obtain the final self-expression matrix $\mathbf{C}$

$$\mathbf{C} = \lambda \mathbf{C}_s + (1-\lambda)\,\mathbf{C}_{sa}. \tag{11}$$

Then, we can construct an informative affinity matrix $\mathbf{A}$ by

$$\mathbf{A} = \frac{1}{2}\left(|\mathbf{C}| + |\mathbf{C}|^{\top}\right). \tag{12}$$

Finally, we apply a spectral clustering algorithm on $\mathbf{A}$ to get the superpixel-level clustering result $\mathbf{Y}_S$. The pixel-level clustering result $\mathbf{Y}$ can be directly obtained by treating the pixels in a superpixel field as one category corresponding to the superpixel-level class.

D. Objective Function and Training

The objective function of SCSC consists of four parts: consistency loss $\mathcal{L}_{\text{cons}}$, reconstruction loss $\mathcal{L}_{\text{rec}}$, superpixel contrastive loss $\mathcal{L}_{\text{cont}}$, and self-expressive loss $\mathcal{L}_{\text{selfexp}}$.

The consistency loss aims to improve the ability of the HMAE encoder to better explore the consistency between two self-expression matrices $\mathbf{C}_s$ and $\mathbf{C}_{sa}$. It can be formulated as follows:

$$\mathcal{L}_{\text{cons}} = \|\mathbf{C}_s - \mathbf{C}_{sa}\|_F^2. \tag{13}$$

The reconstruction loss is the typical DSC loss which aims to extract the spatial-box features in a self-supervised way. In our MSCSC, only the masked patches within each spatial-box need to be reconstructed, whose loss function is formulated as follows:

$$\mathcal{L}_{\text{rec}} = \frac{1}{2Mn_p}\left(\left\|\mathbf{X}_s - \hat{\mathbf{X}}_s\right\|^2 + \left\|\mathbf{X}_{sa} - \hat{\mathbf{X}}_{sa}\right\|^2\right) \tag{14}$$

where M denotes the number of superpixel blocks and n_p denotes the number of masked patches.

For the contrastive loss, we first define the pairwise similarity of z_a and z_b as follows:

$$s(z_a, z_b) = \frac{(z_a)^{\top}(z_b)}{\|z_a\|\,\|z_b\|}. \tag{15}$$

Given a positive pair of samples x_{s_i} and x_{sa_i}, x_{s_i} with the rest of samples $\{x_{s_j}\}_{j\neq i}$ or $\{x_{sa_j}\}_{j\neq i}$ can be seen as negative pairs. We utilize the NT-Xent loss [57] as our superpixel contrastive loss

$$l(x_i, x_j) = -\log \frac{\exp\left(s\left(h_i, h_j\right)/\tau\right)}{\sum_{k=1}^{2M} \mathbf{1}_{[k\neq i]} \exp\left(s\left(h_i, h_j\right)/\tau\right)} \tag{16}$$

$$\mathcal{L}_{\text{cont}} = \frac{1}{2M}\sum_{i=1}^{M} l\left(x_{s_i}, x_{sa_i}\right) \tag{17}$$

where h_i and h_j are the latent features extracted from the weight shared MLP layers, and τ represents the temperature parameter [60], which controls the strength of penalty applied to hard negative samples.

The self-expression loss is designed for optimizing $\mathbf{C}_s$ and $\mathbf{C}_{sa}$ with the latent features reconstruction and sparse regulation

$$\mathcal{L}_{\text{selfexp}} = \frac{1}{2M}\left(\|\mathbf{Z}_v - \mathbf{Z}_v\mathbf{C}_s\|_F^2 + \|\mathbf{Z}_{va} - \mathbf{Z}_{va}\mathbf{C}_{sa}\|_F^2\right) + \|\mathbf{C}_s\|_2 + \|\mathbf{C}_{sa}\|_2. \tag{18}$$

Above all, the final loss function of our proposed method is defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cons}} + \gamma_1\mathcal{L}_{\text{rec}} + \gamma_2\mathcal{L}_{\text{cont}} + \gamma_3\mathcal{L}_{\text{selfexp}} \tag{19}$$

where γ_1,γ_2, and γ_3 are the tradeoff parameters of each part of the loss function.

Algorithm 1 Procedure for Training MSCSC

Input: HSI dataset $\mathbf{X}$; the tradeoff parameters γ_1, γ_2, γ_3; the weight of self-expression λ; temperature parameter τ; the number of superpixel blocks M; maximum iteration T_{max}

Output: Clustering result $\mathbf{Y}$

```
Preprocessing X to obtain spatial-box X_p;
Obtain superpixel segmentation S from X through SLIC;
Pretraining HMAE using X_p, and obtain the pretrained parameters θ_E and θ_D;
Initialize the MSCSC network with θ_E and θ_D, respectively;
while T < T_max do
    sample random augmentation X_s and average augmentation X_sa as the superpixel-level input;
    Calculate self-expression properties C_s and C_sa;
    Update the network parameters to minimize L through Eq. (19);
end
Calculate the fused self-expression coefficient matrix C through Eq. (11);
Calculate the affinity matrix A through Eq. (12);
Obtain the superpixel-level clustering result Y_S by conducting spectral clustering on A;
Project Y_S to pixel-level clustering result Y.
```

We adopt AdamW optimizer to optimize our proposed method. The pretraining epochs are set to 500 until convergence. The pretrained parameters are transferred to initialize the MSCSC network, and the epochs are set to 100. Spectral clustering on $\mathbf{A}$ generates the final clustering results, and the Hungarian algorithm [61] is employed to ensure label indices align with the ground truth. As shown in Algorithm 1, we provide the detailed procedure of MSCSC.

IV. Experiments

In this section, we conduct the comparative experiments on four well-known large-scale HSI datasets to evaluate the performance of our proposed MSCSC, i.e., the Indian Pines dataset, the Salinas dataset, the Pavia Center dataset, and the WHU_Hi_HanChuan dataset.

A. Datasets

1) Indian Pines Dataset: This dataset is among the earliest datasets developed for HSI classification. It was acquired in June 1992 over a pine plantation located in northwestern Indiana, USA, using the Airborne Visible/Infrared Imaging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED LAND-COVER CATEGORIES OF FOUR LARGE-SCALE HSI DATASETS

	Indian Pines		Salinas		Pavia Center		WHU_Hi_HanChuan	
No.	class	sample	class	sample	class	sample	class	sample
1	Alfalfa	46	Brocoli_green_weeds_1	2009	Water	65278	Strawberry	44735
2	Corn-notill	1428	Brocoli_green_weeds_2	3726	Trees	6507	Cowpea	22753
3	Corn-mintill	830	Fallow	1976	Asphalt	2891	Soybean	10287
4	Corn	237	Fallow_rough_plow	1394	Self-Blocking Bricks	2140	Sorghum	5353
5	Grass-pasture	483	Fallow_smooth	2678	Bitumen	6549	Water spinach	1200
6	Grass-trees	730	Stubble	3959	Tiles	7525	Watermelon	4533
7	Grass-pasture-moved	28	Celery	3579	Shadows	7287	Greens	5903
8	Hay-windrowed	478	Grapes_untrained	11271	Meadows	3122	Trees	17978
9	Oats	20	Soil_vinyard_develop	6203	Bare Soil	2165	Grass	9469
10	Soybean-notill	972	Corn_senesced_green_weeds	3278	**Unlabeled**	**433576**	Red roof	10516
11	Soybean-mintill	2455	Lettuce_romaine_4wk	1068			Gray roof	16911
12	Soybean-clean	593	Lettuce_romaine_5wk	1927			Plastic	3679
13	Wheat	205	Lettuce_romaine_6wk	916			Bare soil	9116
14	Woods	1265	Lettuce_romaine_7wk	1070			Road	18560
15	Buildings-Grass-Trees-Drives	386	Vinyard_untrained	7268			Bright object	1136
16	Stone-Steel-Towers	93	Vinyard_vertical_trellis	1807			Water	75401
17	**Unlabeled**	**10776**	**Unlabeled**	**56975**			**Unlabeled**	**111221**

Spectrometer (AVIRIS). The dataset covers a wavelength range of 0.4–2.5 μm with a spatial resolution of approximately 20 $\times$ 20 m, which makes it susceptible to generating mixed pixels, adding challenges to pixel-level classification of ground objects. It consists of 145 $\times$ 145 pixels and 200 continuous spectral bands (excluding 20 water absorption bands), totaling 21025 pixels, with 10249 pixels labeled into 16 categories.

2) Salinas Dataset: This dataset was acquired in the Salinas Valley region of California, USA, using the AVIRIS remote sensor. Salinas dataset has dimensions of 512×217 pixels and includes 204 continuous spectral bands, with 20 bands related to noise and water absorption typically removed. Compared with the Indian Pines dataset, the dataset has a higher spatial resolution. There are totally 111104 pixels in the Salinas dataset, with 54129 pixels labeled into 16 categories.

3) Pavia Center Dataset: This dataset was acquired over the Pavia region in Italy using the Reflective Optics Spectrographic Imaging System (ROSIS). Pavia Center dataset covers a wavelength range of 430–860 nm and consists of 1096×490 pixels across 102 relatively noise-free spectral bands. There are totally 537040 pixels in the Pavia Center dataset, with 103464 of them labeled into nine distinct categories.

4) WHU_Hi_HanChuan Dataset: The Dataset was acquired on June 17, 2016, using the Headwall Nano-Hyperspec sensor on the Leica Aibot X6 UAV in Hanchuan, Hubei, China. The dataset contains 1217 $\times$ 303 pixels and 274 spectral bands (400–1000 nm) with a spatial resolution of about 0.109 m. The dataset includes 257530 labeled samples across 16 categories.

The detailed categories of four large-scale HSI datasets are summarized in Table I. It should be noted that there are many unlabeled samples participating in training, but their labels assigned by the model do not participate in the evaluation process.

B. Experimental Settings

All results of our method and the comparisons were conducted via a NVIDIA TITAN RTX GPU with 24-GB graphics memory. The experimental environment is PyTorch 2.0.0 and Python 3.9.2. We set the principal components $p = 50$, the window size of spatial-box $w = 7$ and the masking ratio $\sigma = 0.75$ on all four datasets. The input spatial-box is of size $7 \times 7 \times 50$, containing only 49 patches of size $1 \times 1 \times 50$. The embedding dimension of encoder and decoder is set to 96 and 64. The layer of encoder and decoder is set to 6 and 2, respectively. As shown in Fig. 3, the number of superpixel blocks set on Indian Pines dataset ($M = 600$), the Salinas dataset ($M = 800$), the Pavia Center dataset ($M = 2000$), and the WHU_Hi_HanChuan dataset ($M = 1800$). The learning rate is set to 0.0005.

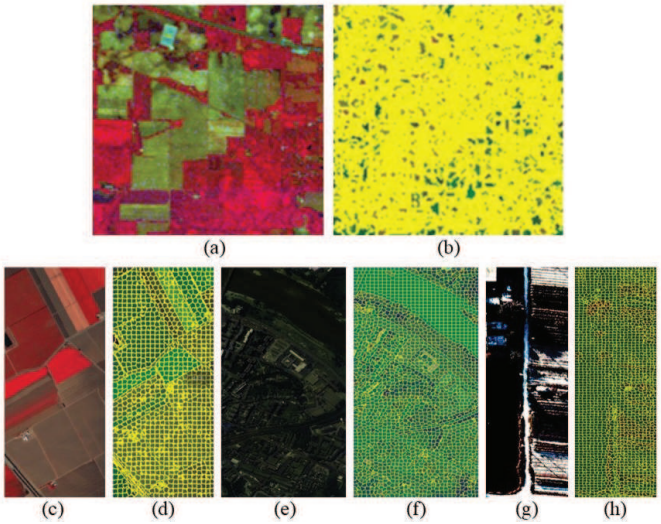


Fig. 3. Visualization of four datasets. (a) False-color image of Indian Pines. (b) Segmentation of Indian Pines. (c) False-color image of Salinas. (d) Segmentation of Salinas. (e) False-color image of Pavia Center. (f) Segmentation of Pavia Center. (g) False-color image of WHU_Hi_HanChuan. (h) Segmentation of WHU_Hi_HanChuan.

1) Comparison Algorithms: Our proposed method is compared against other advanced large-scale clustering approaches, including the centroid-based methods k-means [18], fuzzy c-means (FCM) [20], the traditional subspace clustering methods SSSC [39], S^5C [40], SAPC [62], the deep

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF RESULTS OBTAINED BY VARIOUS SUBSPACE CLUSTERING METHODS ON THE INDIAN PINES DATASET

class	k-means	FCM	SSSC	S^5C	SAPC	SENet	NCSC	SDST	S^2GCL	MSCSC
Alfalfa(%)	0.00	**17.39**	0.00	8.70	0.00	0.00	0.00	0.00	0.00	0.00
Corn-notill(%)	24.51	27.10	21.64	33.05	33.54	33.89	23.39	41.11	43.98	**64.43**
Corn-mintill(%)	40.84	27.23	34.94	32.41	**49.04**	31.81	33.86	40.60	36.39	25.90
Corn(%)	32.49	28.69	33.76	9.28	0.00	6.33	**45.99**	14.35	0.00	0.00
Grass-pasture(%)	49.69	57.14	59.83	62.32	65.01	44.31	**65.63**	52.17	**65.63**	**65.63**
Grass-trees(%)	39.86	29.32	56.44	61.37	77.95	90.55	55.89	79.73	57.40	**96.16**
Grass-pasture-moved(%)	78.57	60.71	46.43	78.57	100	42.86	0.00	3.57	**100**	**100**
Hay-windrowed(%)	89.12	93.57	54.60	55.86	94.35	63.60	95.61	99.16	**100**	**100**
Oats(%)	0.00	10.00	0.00	20.00	0.00	0.00	0.00	35.00	0.00	**100**
Soybean-notill(%)	19.24	30.45	27.06	28.50	30.04	39.81	56.89	20.16	**67.90**	56.28
Soybean-mintill(%)	36.25	36.74	36.01	41.87	34.38	34.22	40.69	73.48	59.02	**78.98**
Soybean-clean(%)	17.03	22.43	10.29	16.69	11.80	14.84	0.00	13.32	**73.52**	53.96
Wheat(%)	98.54	95.12	77.56	83.41	99.02	14.15	**100**	96.59	**100**	98.54
Woods(%)	42.21	45.06	47.75	46.40	70.43	40.79	55.73	63.00	60.16	**69.49**
Buildings-grass-trees-drives(%)	18.39	17.36	25.91	8.55	27.72	17.10	75.39	30.83	76.94	**99.74**
Stone-steel-towers(%)	83.87	47.31	68.82	86.02	0.00	41.94	2.15	**96.77**	0.00	90.32
OA(%)	37.15	37.58	36.97	39.85	45.42	38.24	45.48	54.22	58.35	**68.65**
AA(%)	41.91	40.35	37.56	42.06	43.33	32.26	40.70	47.49	52.56	**68.71**
$kappa$	0.3042	0.3110	0.3043	0.3268	0.3884	0.3267	0.4000	0.4749	0.5400	**0.6418**
NMI	0.4339	0.4394	0.3937	0.4334	0.5108	0.3843	0.5035	0.5168	0.6230	**0.6576**

TABLE III
COMPARISON OF RESULTS OBTAINED BY VARIOUS SUBSPACE CLUSTERING METHODS ON THE SALINAS DATASET

class	k-means	FCM	SSSC	S^5C	SAPC	SENet	NCSC	SDST	S^2GCL	MSCSC
Brocoli_green_weeds_1(%)	99.75	99.80	0.00	0.00	0.00	0.00	0.00	0.00	**100**	**100**
Brocoli_green_weeds_2(%)	39.64	70.45	88.03	97.91	99.57	96.38	**100**	99.89	**100**	**100**
Fallow(%)	0.00	15.84	0.00	44.53	0.00	62.15	77.02	35.24	**100**	**100**
Fallow_rough_plow(%)	**99.28**	71.74	0.00	77.33	87.88	98.92	0.00	94.40	0.00	0.00
Fallow_smooth(%)	95.00	99.22	**99.96**	77.04	97.50	91.82	99.85	96.49	98.92	99.85
Stubble(%)	94.90	98.96	81.01	99.47	99.12	99.19	99.70	98.61	**100**	99.82
Celery(%)	98.24	0.00	0.64	27.94	98.97	99.47	**100**	99.22	99.56	99.97
Grapes_untrained(%)	62.24	42.24	82.49	53.06	81.15	63.99	**92.79**	63.31	63.75	82.01
Soil_vinyard_develop(%)	0.49	98.10	96.15	58.70	98.26	88.76	**99.97**	97.40	98.48	99.94
Corn_sensced_green_weeds(%)	90.50	63.30	45.03	24.53	66.87	60.01	71.90	65.01	**93.29**	92.40
Lettuce_romaine_4wk(%)	74.53	0.00	0.00	56.09	**95.88**	85.49	52.53	0.00	0.00	93.26
Lettuce_romaine_5wk(%)	97.41	36.33	71.77	67.51	94.14	98.60	0.00	**100**	95.54	0.00
Lettuce_romaine_6wk(%)	**98.69**	97.05	18.67	98.14	97.16	0.00	0.00	98.58	0.00	0.00
Lettuce_romaine_7wk(%)	88.50	81.21	63.36	88.88	1.02	93.46	0.00	88.22	0.00	**96.17**
Vinyard_untrained(%)	59.82	49.09	16.22	42.05	64.91	53.49	74.13	66.95	**85.20**	84.95
Vinyard_vertical_trellis(%)	0.00	42.22	72.27	65.08	97.18	67.63	**100**	72.27	**100**	96.35
OA(%)	66.88	56.41	56.61	53.99	78.81	71.59	78.00	75.80	81.42	**85.58**
AA(%)	68.69	53.62	45.97	56.89	73.73	66.62	60.49	73.47	70.92	**77.79**
$kappa$	0.6318	0.5209	0.5088	0.4939	0.7637	0.6831	0.7532	0.7313	0.7936	**0.8395**
NMI	0.7235	0.6646	0.6322	0.5995	0.8464	0.7651	0.8257	0.7839	0.8669	**0.8770**

learning methods SENet [47], NCSC [44], SDST [36], and S^2GCL [37].

2) Evaluation Indicators: To evaluate the clustering effectiveness of various unsupervised HSI classification models, we use four metrics: overall accuracy (OA), average accuracy (AA), kappa, and normalized mutual information (NMI). OA measures the proportion of correctly classified pixels, AA calculates the AA across categories, kappa assesses the agreement between true labels and classification results, and NMI measures the similarity between clustering results. For each metric, a higher score indicates better clustering performance.

C. Comparative Experimental Results and Analysis

The quantitative results of the comparison experiments on the four large-scale HSI datasets are listed in Tables II–V. The best results in each row are highlighted in bold. OOM represents out of memory, and there is no relevant result for some algorithms in Tables IV and V.

1) Quantitative Analysis of Indian Pines Dataset: The dataset is relatively small with labeled pixels accounting for about 50% of the total. Clustering this dataset is challenging due to numerous mixed pixels and the similar spectral features of many categories. Table II shows that MSCSC has the highest accuracy and has a significant improvement. It can be seen that MSCSC achieves the highest accuracy, showing significant improvement. Among the classic methods, *k*-means and FCM, FCM has the highest accuracy, indicating that FCM has a better ability to distinguish between mixed pixels and similar spectral features. SSSC and S^5C, traditional two-stage subspace clustering methods, have lower accuracy due to their limited sample selection for subspace clustering. SAPC introduces graph-based technique to improve accuracy, but it heavily relies on the quality of superpixel segmentation. Poor

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF RESULTS OBTAINED BY VARIOUS SUBSPACE CLUSTERING METHODS ON THE PAVIA CENTER DATASET

class	k-means	FCM	SSSC	S^5C	SAPC	SENet	NCSC	SDST	S^2GCL	MSCSC
Water(%)	99.64	**100**	96.03	84.67	**100**	OOM	OOM	OOM	99.36	**100**
Trees(%)	70.94	**100**	65.65	89.90	34.64	–	–	–	50.41	79.36
Asphalt(%)	12.76	0.00	0.00	**48.05**	0.00	–	–	–	38.15	29.12
Self-blocking bricks(%)	0.00	0.00	0.09	0.05	0.00	–	–	–	36.59	**100**
Bitumen(%)	**87.11**	18.08	35.47	58.25	62.42	–	–	–	69,13	71.75
Tiles(%)	93.81	0.00	11.63	99.40	**99.89**	–	–	–	21.51	39.65
Shadows(%)	54.89	0.00	**99.19**	84.19	76.99	–	–	–	68.67	82.12
Meadows(%)	0.00	74.60	**99.97**	99.90	60.76	–	–	–	26.23	94.55
Bare soil(%)	0.00	0.32	45.03	47.39	98.71	–	–	–	48.59	**90.25**
OA(%)	80.02	72.78	78.76	81.27	85.81	–	–	–	80.26	**88.92**
AA(%)	40.48	32.56	50.34	67.98	59.27	–	–	–	50.96	**76.31**
$kappa$	0.6464	0.4218	0.6331	0.7066	0.7556	–	–	–	0.6617	**0.8096**
NMI	0.7405	0.4992	0.6401	0.7437	**0.8320**	–	–	–	0.6594	0.8157

Note: OOM represents out of memory

TABLE V
COMPARISON OF RESULTS OBTAINED BY VARIOUS SUBSPACE CLUSTERING METHODS ON THE WHU_HI_HANCHUAN DATASET

class	k-means	FCM	SSSC	S^5C	SAPC	SENet	NCSC	SDST	S^2GCL	MSCSC
Strawberry(%)	37.75	31.20	48.81	39.43	56.61	OOM	OOM	OOM	46.94	**97.18**
Cowpea(%)	17.95	14.46	48.42	44.46	46.80	–	–	–	30.85	**77.63**
Soybean(%)	29.38	24.26	69.29	80.41	68.90	–	–	–	81.05	**96.65**
Sorghum(%)	42.69	35.08	93.46	94.04	93.69	–	–	–	0.00	**97.80**
Water spinach(%)	28.67	**40.83**	0.00	0.00	0.00	–	–	–	0.00	0.00
Watermelon(%)	15.40	13.92	0.99	0.00	1.39	–	–	–	**100**	0.00
Greens(%)	47.45	68.12	87.31	82.30	0.00	–	–	–	**98.92**	0.00
Trees(%)	18.37	16.20	38.47	13.35	39.79	–	–	–	40.84	**79.87**
Grass(%)	8.65	7.69	32.91	36.89	17.36	–	–	–	**51.33**	44.13
Red roof(%)	20.78	10.76	57.67	0.00	42.61	–	–	–	**80.06**	31.66
Gray roof(%)	75.08	74.90	56.81	73.15	62.86	–	–	–	51.37	**83.28**
Plastic(%)	0.00	0.00	16.09	0.00	19.46	–	–	–	**85.10**	0.00
Bare soil(%)	10.94	11.91	0.00	**56.81**	0.11	–	–	–	46.30	9.23
Road(%)	14.05	20.64	**42.26**	37.40	42.18	–	–	–	41.17	34.88
Bright object(%)	33.01	53.43	11.53	39.96	**56.69**	–	–	–	32.31	0.00
Water(%)	70.11	70.21	36.71	54.92	48.52	–	–	–	72.12	**99.76**
OA(%)	41.15	39.86	43.54	45.89	45.75	–	–	–	56.60	**75.64**
AA(%)	29.39	30.84	40.05	40.82	37.31	–	–	–	**53.65**	47.00
$kappa$	0.3218	0.3145	0.3874	0.4043	0.3988	–	–	–	0.5168	**0.7099**
NMI	0.3859	0.3862	0.5659	0.5577	0.5555	–	–	–	0.6571	**0.7145**

Note: OOM represents out of memory

superpixel segmentation may generate low-quality anchors, impacting clustering performance. SENet, NCSC, SDST, and S^2GCL are all deep clustering methods for large datasets. Although they use deep neural network to extract features, their clustering performance is still suboptimal. Compared with these methods, MSCSC achieves over 60% OA for the first time, outperforming S^2GCL by 10.30%, 16.15%, 0.1018, and 0.0346 on OA, AA, kappa, and NMI, respectively, demonstrating the robustness and effectiveness of MSCSC.

2) Quantitative Analysis of Salinas Dataset: Due to the relatively pure spectral features of this dataset, the classification accuracy is significantly higher than Indian Pines dataset. Table III shows the detailed clustering performance of each comparison method. It can be seen that *k*-means performs best among the classic clustering algorithms, with OA being 10.47% higher than FCM. This indicates that *k*-means has better clustering performance when the spectral bands are distinguishable. Although SSSC achieved a better performance in the Fallow smooth category, it performed poorly in OA compared with S^5C. Both graph-based and deep clustering methods achieved the best accuracy in multiple categories, with OA approaching 80%. Specifically, the S^2GCL method achieved an OA of 81.42% and reached the best performance in seven subcategories. Compared with this, our method outperforms in OA, AA, kappa, and NMI by 4.16%, 6.87%, 0.0459, and 0.0101, respectively.

3) Quantitative Analysis of Pavia Center Dataset: The data volume is huge, and labeled pixels account for approximately 20% of the total pixels. The detailed quantification of the clustering performance for each comparison method is shown in Table IV. It is worth noting that the three deep clustering methods SENet, NCSC, and SDST are unable to work on such large-scale HSI datasets due to the large memory consumption. Notably, although S^2GCL performs well, there remains some room for improvement. Our proposed MSCSC benefits from the lightweight HMAE backbone and SCSC network, enabling

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

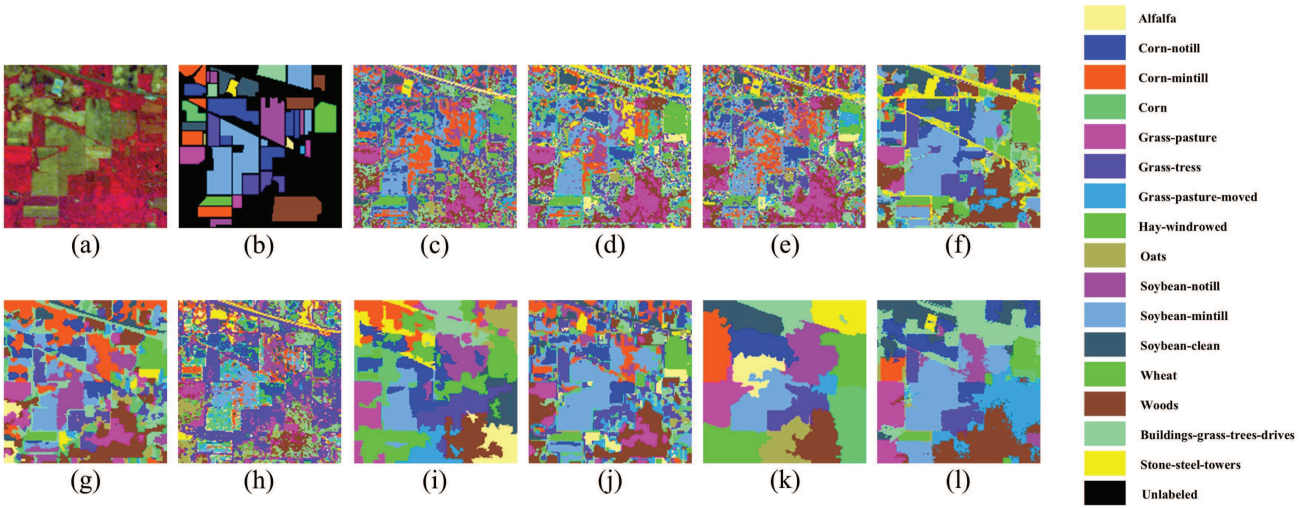


Fig. 4. Clustering visualization of different subspace clustering methods on the Indian Pines dataset. (a) False-color composite image. (b) Label. (c) *k*-means. (d) FCM. (e) SSSC. (f) S^5C. (g) SAPC. (h) SENet. (i) NCSC. (j) SDST. (k) S^2GCL. (l) MSCSC.

it to perform effectively on large-scale datasets. It outperforms the current SOTA methods in OA, AA, and kappa. It is noteworthy that MSCSC performs relatively poorly in AA, which is 12.61% lower than OA. The main reason is that the introduction of superpixel segmentation makes it difficult to correctly clustering the categories with a small sample size. Overall, MSCSC can achieve good performance on large-scale HSI datasets, which is sufficient to demonstrate its superiority.

4) Quantitative Analysis of WHU_Hi_HanChuan Dataset: Due to the similarity in land-cover types and the close spectral characteristics of the HanChuan dataset, misclassification is prone to occur, resulting in relatively low clustering accuracy for many methods on this dataset. As shown in Table V, among the four deep clustering methods, SENet, NCSC, and SDST are also faced with out-of-memory problem as their performances on the Pavia Center dataset. In contrast, S^2GCL outperformed traditional clustering methods like *k*-means, showing that deep clustering method can extract higher level semantic relationships and better integrate spectral–spatial features. This allows for better modeling of the relationships between the original data. Our proposed MSCSC achieves the best performance, particularly in the OA, kappa, and NMI indicators, with improvements of 19.04%, 0.1931, and 0.0574, respectively, compared with the suboptimal method. Although the AA failed to achieve the best result, the gap with the second-best method remained relatively small. This performance is due to the effective combination of the HMAE backbone and SCSC network, reducing interference from nontarget regions, capturing intrinsic HSI features, and minimizing misclassification, showing excellent adaptability across datasets.

D. Qualitative Analysis

As observed from Figs. 4–7, pixel-level methods (such as FCM, SSSC, S^5C, and SENet) exhibit significant salt and pepper noise in their classification maps, failing to accurately cluster samples within contiguous regions that should belong to the same category. This phenomenon primarily stems from the fact that pixel-level methods treat samples as isolated entities, ignoring the spatial information inherent in HSI. In contrast, the clustering results obtained by superpixel-level methods demonstrate smoother and more homogeneous class regions, highlighting the superiority of superpixel approaches in HSI clustering tasks. It is noted that the number of superpixel blocks significantly influences the final clustering results. In S^2GCL, although the quantitative results achieve suboptimal performance across four datasets, the relatively small number of superpixel blocks results in larger block sizes, thereby compromising the precision of ground object segmentation. In comparison, our proposed method employs a larger number of superpixel blocks, not only demonstrating excellent performance in major class segmentation but also achieving precise boundary delineation. This improvement can be primarily attributed to the HMAE's capability of extracting intrinsic features from HSI, making it more sensitive to the boundaries between target and nontarget regions. Taking the Pavia Center dataset as an example, our method achieves accurate clustering partitions for categories containing a substantial number of samples, with boundaries clearly defined and well-aligned with the ground truth labels. Similarly, for the WHU_Hi_HanChuan dataset, our method demonstrates superior performance in clustering maps for various categories, including Strawberry, Cowpea, Soybean, and Sorghum, effectively capturing the distinctive characteristics of each class.

E. Parameters Analysis

Our proposed MSCSC totally includes seven important parameters that significantly impact clustering performance. We conduct many experiments to analyze these parameters.

1) Parameters γ_1, γ_2 and γ_3: In this study, we focus on analyzing the impact of γ_2 and γ_3 on the OA of the proposed MSCSC method, with γ_1 fixed at its optimal value for each dataset. Specifically, we set $\gamma_1 \in \{1, 10, 100, 1000, 10000\}$ and $\gamma_2 \in \{0.1, 1, 10, 100, 1000, 10000\}$, and $\gamma_3 \in$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

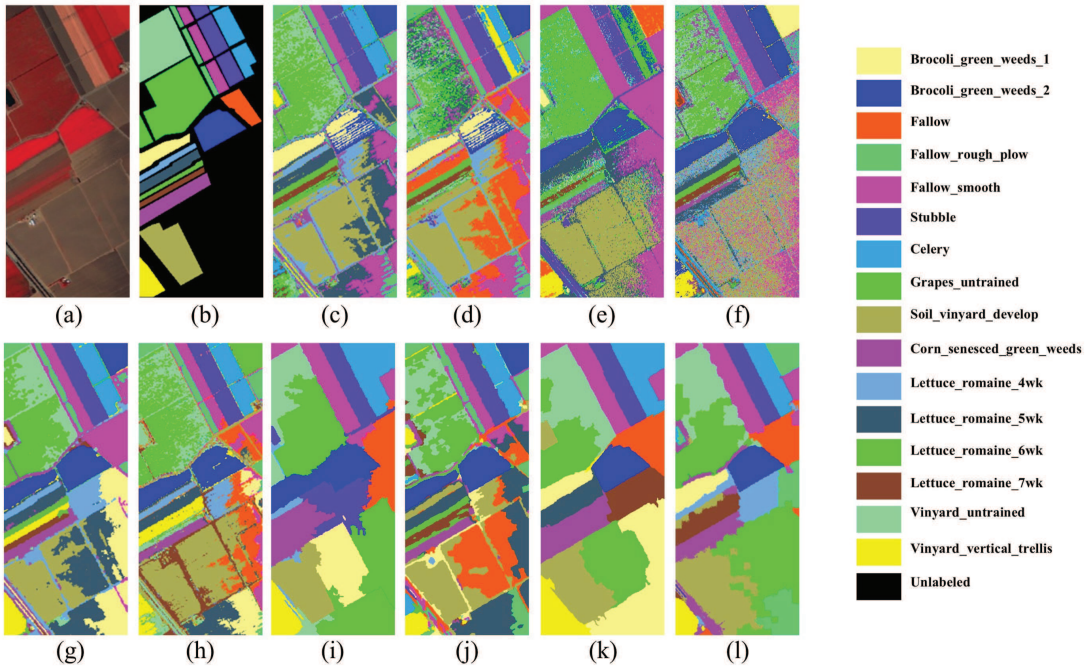


Fig. 5. Clustering visualization of different subspace clustering methods on the Salinas dataset. (a) False-color composite image. (b) Label. (c) k-means. (d) FCM. (e) SSSC. (f) S^5C. (g) SAPC. (h) SENet. (i) NCSC. (j) SDST. (k) S^2GCL. (l) MSCSC.

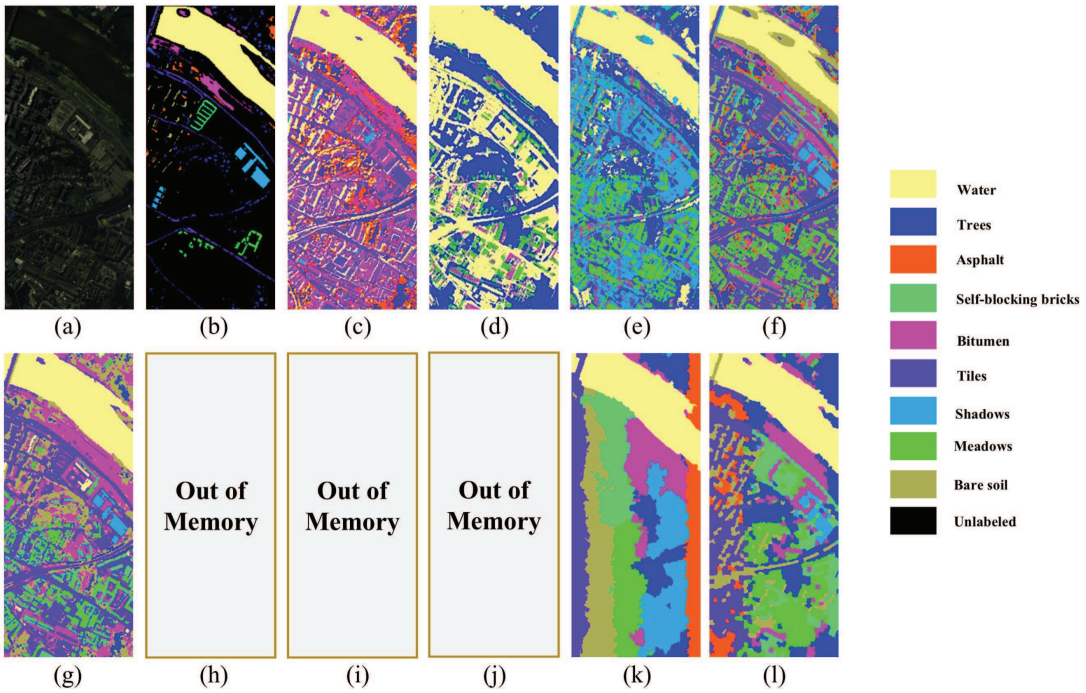


Fig. 6. Clustering visualization of different subspace clustering methods on the Pavia Center dataset. (a) False-color composite image. (b) Label. (c) k-means. (d) FCM. (e) SSSC. (f) S^5C. (g) SAPC. (h) SENet. (i) NCSC. (j) SDST. (k) S^2GCL. (l) MSCSC.

$\{0.1, 1, 10, 100, 1000, 10000\}$. Due to space limitations, the optimal γ_1 values were determined for each dataset: Indian Pines ($\gamma_1 = 1000$), Salinas ($\gamma_1 = 1000$), Pavia Center ($\gamma_1 = 1000$), and WHU_Hi_HanChuan ($\gamma_1 = 10000$). The changing trend of MSCSC in OA in relation to each tradeoff parameter across four datasets is shown in Fig. 8. These results highlight that our method achieves robust performance across a broad range of parameter settings.

2) Parameter w: The window size of spatial-box w is a critical parameter that relates how much spatial–spectral information each sample contains. As shown in Table VI, the experimental results demonstrate that $w = 7 \times 7$ consistently achieves the best performance in terms of OA, AA, kappa, and NMI. While larger windows (e.g., 9×9 and 11×11) occasionally improved accuracy, they also increased computational complexity without providing substantial performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

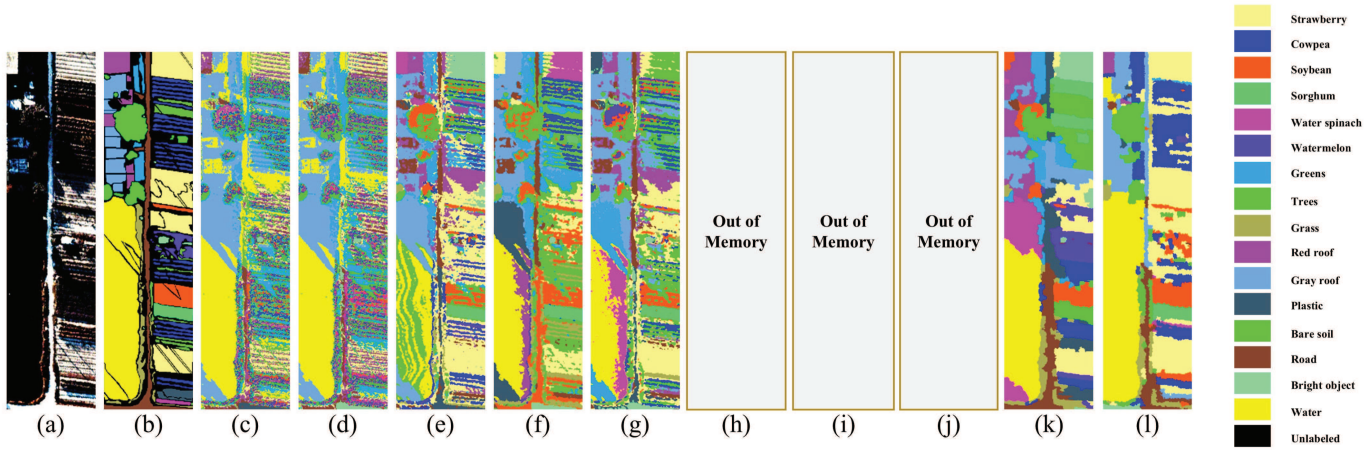


Fig. 7. Clustering visualization of different subspace clustering methods on the WHU_Hi_HanChuan dataset. (a) False-color composite image. (b) Label. (c) k-means. (d) FCM. (e) SSSC. (f) S^5C. (g) SAPC. (h) SENet. (i) NCSC. (j) SDST. (k) S^2GCL. (l) MSCSC.

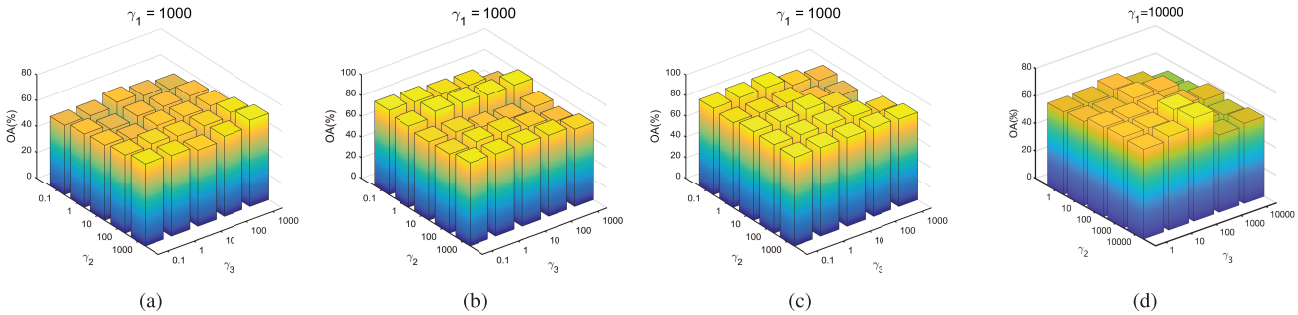


Fig. 8. OA comparison of different γ_2 and γ_3 for the proposed MSCSC on four datasets, with γ_1 fixed at its optimal value. (a) Indian Pines dataset. (b) Salinas dataset. (c) Pavia Center dataset. (d) WHU_Hi_HanChuan dataset.

TABLE VI
CLUSTERING PERFORMANCE OF A DIFFERENT SPATIAL-BOX

Dataset	The size of w	OA(%)	AA(%)	$kappa$	NMI
Indian Pines	5×5	52.56	34.10	0.4343	0.5519
	7×7	**68.65**	**68.71**	**0.6418**	**0.6576**
	9×9	65.38	59.71	0.6032	0.6300
	11×11	59.07	53.79	0.5291	0.6093
Salinas	5×5	78.37	71.74	0.7557	0.8817
	7×7	**85.58**	77.79	**0.8395**	0.8770
	9×9	81.60	**88.51**	0.7930	**0.9073**
	11×11	83.99	76.85	0.8209	0.8727
Pavia Center	5×5	85.94	62.73	0.7585	0.7798
	7×7	**88.92**	**76.31**	**0.8096**	**0.8157**
	9×9	88.69	76.29	0.8052	0.8114
	11×11	88.56	76.04	0.8037	0.8114
WHU_Hi_HanChuan	5×5	68.38	41.62	0.6237	0.6648
	7×7	**75.64**	**47.00**	**0.7099**	**0.7145**
	9×9	73.20	40.67	0.6812	0.6955
	11×11	71.90	40.56	0.6660	0.6861

gains. Conversely, smaller windows (e.g., 5×5) failed to capture sufficient spatial context, leading to lower clustering accuracy. Balancing computational efficiency and clustering performance, we selected the 7×7 window as the optimal size for our experiments, as it effectively captures spatial information while maintaining reasonable computational overhead.

3) Parameter τ: Temperature τ acts as a parameter that adjusts the severity of penalties on challenging negative samples. In this experiment, we fix other parameters and set

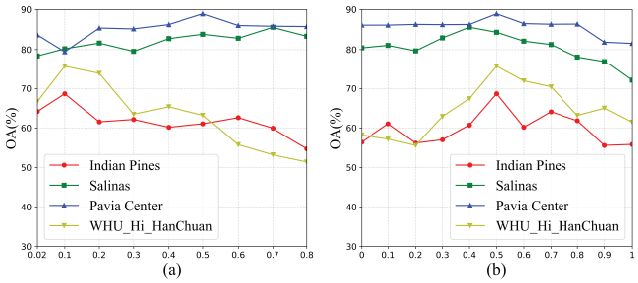


Fig. 9. OA comparison under different parameters of the proposed MSCSC on four large-scale datasets. (a) Temperature parameter τ. (b) Self-expression weight parameter λ.

$\tau \in \{0.02, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ to explore the impact of τ on four datasets. As shown in Fig. 9(a), for the Indian Pines and WHU_Hi_HanChuan datasets, smaller values of τ generally lead to higher clustering performance. In contrast, for the Salinas and Pavia Center datasets, larger values of τ tend to yield better clustering results. This trend can be clearly observed in the figure. Based on these observations, we set $\tau = 0.1$ for the Indian Pines dataset and the WHU_Hi_HanChuan dataset, $\tau = 0.7$ for the Salinas dataset, $\tau = 0.5$ for the Pavia Center dataset.

4) Parameter λ: The self-expression weight parameter λ affects the quality of the fusion self-expression matrix $\mathbf{C}$. we set $\lambda \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$ with other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

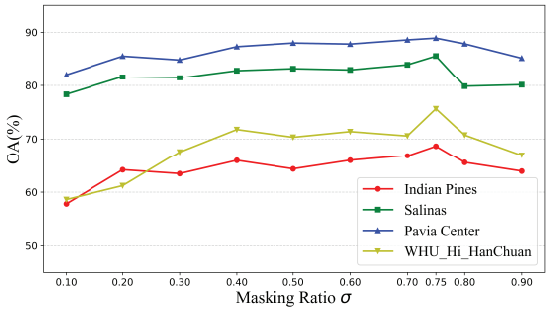


Fig. 10. OA comparison under varying masking ratios of the proposed MSCSC on four large-scale datasets.

TABLE VII
ABLATION STUDY ON FOUR LARGE-SCALE HSI DATASETS

	MSCSC-v1	MSCSC-v2	MSCSC-v3	MSCSC
Pretrained weight of autoencoder	✔	✔	-	✔
CAE backbone	-	✔	-	-
HMAE backbone	✔	-	✔	✔
Dual-branch contrastive subspace clustering	-	✔	✔	✔
Indian Pines	56.26	60.91	40.75	**68.65**
Salinas	81.28	81.43	80.17	**85.58**
Pavia Center	80.21	86.04	84.06	**88.92**
WHU_Hi_HanChuan	59.17	67.31	63.85	**75.64**

parameters fixed. $\lambda = 0$ means that constructing the affinity matrix only by $\mathbf{C}_{sa}$, and $\lambda = 1$ means that constructing the affinity matrix only by $\mathbf{C}_s$. As depicted in Fig. 9(b), the clustering accuracy reaches peak as λ close to median. The main reason is that $\mathbf{C}_s$ and $\mathbf{C}_{sa}$ contain the self-expression information at a different angle, and their integration can make $\mathbf{C}$ more informative. This experiment demonstrates that our strategy of fusing two self-expression matrices is effective.

5) Masking Ratio σ: To comprehensively evaluate the effectiveness of the proposed method, we further investigate the impact of masking ratio σ within the HMAE on four datasets. As shown in Fig. 10, we evaluate σ ranging from 0.1 to 0.9. It can be observed that the optimal clustering performance generally emerged within the 0.6–0.8 masking ratio σ. Although the four datasets exhibit varying levels of sensitivity to the masking ratio, all of them achieve the best performance at $\sigma = 0.75$. The result demonstrates that our HMAE is capable of effectively capturing the intrinsic homogeneity of HSI through appropriate masked reconstruction, while simultaneously suppressing the influence of nontarget region, which contributes to the better clustering performance.

F. Ablation Study

To validate the effectiveness of the proposed MSCSC framework, we conducted an ablation study on four large-scale HSI datasets. As shown in Table VII, the study compared the full MSCSC framework, which employs the proposed HMAE backbone and dual-branch contrastive subspace clustering, against three variants: MSCSC-v1, which retains the HMAE backbone but replaces the dual-branch contrastive subspace clustering with a single branch; MSCSC-v2, which replaces the HMAE backbone with a traditional CAE; and MSCSC-v3, which uses an untrained HMAE backbone approach. The results demonstrate that the full MSCSC framework consistently outperforms all variants across all datasets, achieving the highest OA. This highlights the critical contributions of the HMAE backbone and the dual-branch contrastive subspace clustering mechanism to the framework's performance. Specifically, the HMAE backbone, with its pretrained weights, significantly enhances feature extraction, while the SCSC network improves subspace clustering by leveraging complementary information, which confirms the necessity and effectiveness of these components in our proposed method.

G. Clustering Visualization

To validate the clustering performance of MSCSC, we performed t-distributed stochastic neighbor embedding (t-SNE) [63]. Fig. 11(a)–(d) denotes the original distribution of average superpixel-level features, while Fig. 11(e)–(h) denotes the distributions after HMAE and SCSC. In the original distribution, the categories are more dispersed, especially in the Pavia Center and WHU_Hi_HanChuan datasets, which validates that the discriminability of features based on averaged superpixels is poor. After feature extraction using HMAE and joint optimization of the SCSC network, the intraclass compactness increases, and the interclass separation becomes more pronounced, presenting a more discriminative distribution. This clearly demonstrates that the proposed method is highly effective in enhancing feature discriminability. From Fig. 12, we can observe that the heatmaps on the four datasets exhibit a distinct block diagonal structure. This indicates that after applying MSCSC, each superpixel sample has learned higher intraclass self-representations and lower interclass self-representations. As a result, a significant improvement in accuracy is achieved in the subsequent spectral clustering.

H. Comparison of Running Time

Table VIII displays the runtime comparison of various methods applied to four large-scale HSI datasets, where OOM means out of memory. It can be obviously seen that the classic clustering methods (e.g., k-means and SSSC) demonstrate notable efficiency but at the cost of reduced accuracy, while pixel-level deep clustering approaches (SENet, NCSC, and SDST) face prohibitive memory demands, resulting in out-of-memory failures on large datasets (Pavia Center and WHU_Hi_HanChuan) and excessively long runtime (up to 7344.62 s). In contrast, S^2GCL and our proposed MSCSC resolve these limitations by compressing input with only superpixel-level input, achieving memory-efficient operation (no OOM errors) with competitive speeds. Specifically, S^2GCL generates anchors using a simple MLP and reduces the number of nodes through superpixel segmentation, effectively lowering computational load. However, due to the model not fully mitigating the impact of intersuperpixel noise, the clustering performance does not achieve optimal results. Our proposed MSCSC notably attains state-of-the-art accuracy while reducing runtime by over 80% compared with conventional deep clustering methods, demonstrating that MSCSC effectively balances computational scalability and clustering precision for large-scale HSI datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

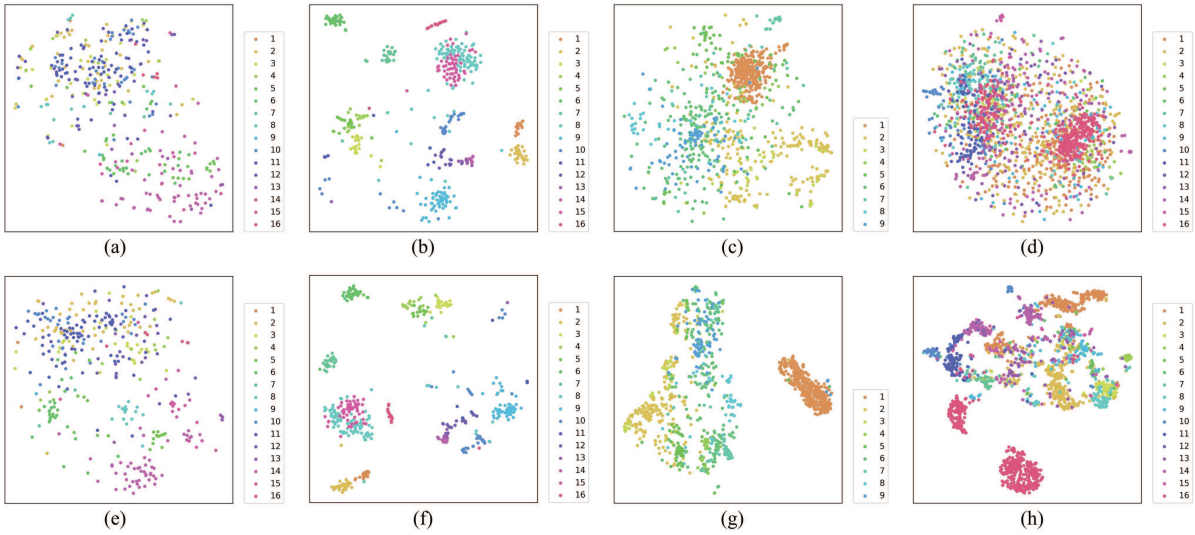


Fig. 11. t-SNE visualization of MSCSC on (a) and (e) Indian Pines, (b) and (f) Salinas, (c) and (g) Pavia Center, and (d) and (h) WHU_Hi_HanChuan.

TABLE VIII
EXECUTION TIME (SECONDS) OF DIFFERENT CLUSTERING METHODS ON FOUR LARGE-SCALE DATASETS

Dataset	k-means	FCM	SSSC	S^5C	SAPC	SENet	NCSC	SDST	S^2GCL	MSCSC
Indian Pines	7.82	135.21	5.24	307.01	5.47	658.11	302.55	1374.05	7.77	59.96
Salinas	21.64	1006.57	40.06	1343.40	12.97	1382.75	944.60	7344.62	14.99	167.48
Pavia Center	61.03	1769.30	122.91	5298.25	93.27	OOM	OOM	OOM	37.35	186.67
WHU_Hi_HanChuan	106.18	3426.05	116.32	4903.68	110.48	OOM	OOM	OOM	27.20	163.52

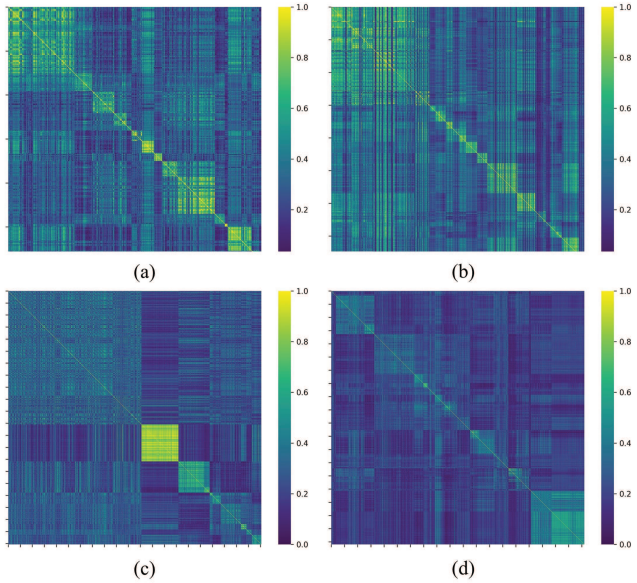


Fig. 12. Self-expression visualization of the proposed MSCSC on (a) Indian Pines, (b) Salinas, (c) Pavia Center, and (d) WHU_Hi_HanChuan.

V. CONCLUSION

In this article, we propose a novel DSC framework named MSCSC for handling large-scale HSI. We design a lightweight MAE-based backbone named HMAE to pretrain the ViT encoder and decoder, which effectively mitigates interference from nontarget region samples in HSI while capturing their intrinsic spatial–spectral features. Then, inspired by CL, we propose SCSC network to learn more discriminative superpixel-level self-representation. Furthermore, we design a novel data augmentation strategy for SCSC to better learn the global and local information of HSI. Experiments on four large-scale HSI datasets demonstrate that MSCSC attains superior clustering results compared with advanced large-scale HSI clustering techniques.

In our experiments, we observed that when the target region occupies a relatively small proportion of the overall area, the superpixel blocks often overlap with the boundaries of the target region. This phenomenon can introduce ambiguity at the target class boundaries, thereby affecting clustering accuracy. In the future, we aim to explore integrating superpixel segmentation into an iterative optimization framework within a network architecture, rather than relying solely on one-time segmentation results. This approach holds the potential to further enhance the model's clustering performance.

REFERENCES

[1] H. Erives and G. J. Fitzgerald, "Automated registration of hyperspectral images for precision agriculture," *Comput. Electron. Agricult.*, vol. 47, no. 2, pp. 103–119, May 2005.

[2] B. Jia et al., "Essential processing methods of hyperspectral images of agricultural and food products," *Chemometric Intell. Lab. Syst.*, vol. 198, Mar. 2020, Art. no. 103936.

[3] R. J. Ellis and P. W. Scott, "Evaluation of hyperspectral remote sensing as a means of environmental monitoring in the St. Austell China clay (kaolin) region, cornwall, U.K.," *Remote Sens. Environ.*, vol. 93, nos. 1–2, pp. 118–130, Oct. 2004.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

[4] M. Yasir, S. Liu, X. Mingming, J. Wan, S. Pirasteh, and K. B. Dang, "ShipGeoNet: SAR image-based geometric feature extraction of ships using convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[5] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.

[6] J. Fan, N. Zhou, J. Peng, and L. Gao, "Hierarchical learning of tree classifiers for large-scale plant species identification," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4172–4184, Nov. 2015.

[7] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[8] M. Li, Y. Fu, T. Zhang, and G. Wen, "Supervise-assisted self-supervised deep-learning method for hyperspectral image restoration," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 1–14, Apr. 2025.

[9] Z. Zhong, J. Li, D. A. Clausi, and A. Wong, "Generative adversarial networks and conditional random fields for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 50, no. 7, pp. 3318–3329, Jul. 2020.

[10] J. Feng, H. Yu, L. Wang, X. Cao, X. Zhang, and L. Jiao, "Classification of hyperspectral images based on multiclass spatial–spectral generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5329–5343, Aug. 2019.

[11] H. Sun, X. Zheng, and X. Lu, "A supervised segmentation network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 2810–2825, 2021.

[12] Z. Gong, W. Hu, X. Du, P. Zhong, and P. Hu, "Deep manifold embedding for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10430–10443, Oct. 2022.

[13] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[14] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[15] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Feb. 2024, Art. no. 5508416.

[16] F. Xu, G. Zhang, C. Song, H. Wang, and S. Mei, "Multiscale and cross-level attention learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501615.

[17] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Jun. 2024, Art. no. 5522214.

[18] T. Kanungo, D. M. Mount, N. S. Netanyahu, C. D. Piatko, R. Silverman, and A. Y. Wu, "An efficient k-means clustering algorithm: Analysis and implementation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 881–892, Jul. 2002.

[19] S. Chen and D. Zhang, "Robust image segmentation using FCM with spatial constraints based on new kernel-induced distance measure," *IEEE Trans. Syst. Man Cybern. B, Cybern.*, vol. 34, no. 4, pp. 1907–1916, Aug. 2004.

[20] J. C. Bezdek, R. Ehrlich, and W. Full, "FCM: The fuzzy c-means clustering algorithm," *Comput. Geosci.*, vol. 10, nos. 2–3, pp. 191–203, Jan. 1984.

[21] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[22] S. Huang, H. Zhang, and A. Pižurica, "Joint sparsity based sparse subspace clustering for hyperspectral images," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3878–3882.

[23] L. Wang et al., "Fast high-order sparse subspace clustering with cumulative MRF for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 152–156, Jan. 2021.

[24] C. Tang, X. Liu, E. Zhu, L. Wang, and A. Zomaya, "Hyperspectral band selection via spatial–spectral weighted region-wise multiple graph fusion-based spectral clustering," in *Proc. 13th Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 3038–3044.

[25] C. Tang et al., "Spatial and spectral structure preserved self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Nov. 2023, Art. no. 5531413.

[26] G. Jiang, J. Zhang, Y. Zhang, X. Jiang, and Z. Cai, "Structured-anchor projected clustering for hyperspectral images," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[27] P. Zhou, Y. Hou, and J. Feng, "Deep adversarial subspace clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1596–1604.

[28] J. Zhang et al., "Self-supervised convolutional subspace clustering network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5468–5477.

[29] M. Kheirandishfard, F. Zohrizadeh, and F. Kamangar, "Deep low-rank subspace clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 864–865.

[30] P. Ji, T. Zhang, H. Li, M. Salzmann, and I. Reid, "Deep subspace clustering networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 30, 2017, pp. 24–33.

[31] M. Zeng, Y. Cai, X. Liu, Z. Cai, and X. Li, "Spectral–spatial clustering of hyperspectral image based on Laplacian regularized deep subspace clustering," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2694–2697.

[32] K. Li, Y. Qin, Q. Ling, Y. Wang, Z. Lin, and W. An, "Self-supervised deep subspace clustering for hyperspectral images with adaptive self-expressive coefficient matrix initialization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3215–3227, 2021.

[33] Z. Zhang, Y. Cai, W. Gong, P. Ghamisi, X. Liu, and R. Gloaguen, "Hypergraph convolutional subspace clustering with multihop aggregation for hyperspectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 676–686, 2022.

[34] T. Li, Y. Cai, Y. Zhang, Z. Cai, and X. Liu, "Deep mutual information subspace clustering network for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] T. Han, S. Niu, X. Gao, W. Yu, N. Cui, and J. Dong, "Deep low-rank graph convolutional subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2025, Art. no. 5533513.

[36] F. Luo, Y. Liu, Y. Duan, T. Guo, L. Zhang, and B. Du, "SDST: Self-supervised double-structure transformer for hyperspectral images clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Mar. 2024, Art. no. 5511614.

[37] A. Yang, M. Li, Y. Ding, X. Xiao, and Y. He, "An efficient and lightweight spectral–spatial feature graph contrastive learning framework for hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Nov. 2024, Art. no. 5537714.

[38] M. E. Wall, A. Rechtsteiner, and L. M. Rocha, "Singular value decomposition and principal component analysis," in *A Practical Approach to Microarray Data Analysis*. Boston, MA, USA: Springer, 2005, pp. 91–109.

[39] X. Peng, L. Zhang, and Z. Yi, "Scalable sparse subspace clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 430–437.

[40] S. Matsushima and M. Brbic, "Selective sampling-based scalable sparse subspace clustering," in *Proc. 33rd Conf. Neural Inf. Process. Syst.*, 2019, pp. 12416–12425.

[41] S. Huang, H. Zeng, H. Chen, and H. Zhang, "Spatial and cluster structural prior guided subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[42] N. Huang, L. Xiao, Y. Xu, and J. Chanussot, "A bipartite graph partition-based coclustering approach with graph nonnegative matrix factorization for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2022, Art. no. 5506918.

[43] N. Huang, L. Xiao, J. Liu, and J. Chanussot, "Graph convolutional sparse subspace coclustering with nonnegative orthogonal factorization for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5512016.

[44] Y. Cai et al., "Superpixel contracted neighborhood contrastive subspace clustering network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530113.

[45] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[46] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.

[47] S. Zhang, C. You, R. Vidal, and C.-G. Li, "Learning a self-expressive network for subspace clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12388–12398.
[48] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[49] E. Elhamifar and R. Vidal, "Sparse subspace clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 2790–2797.
[50] Y. Tian et al., "Unsupervised anomaly detection in medical images with a memory-augmented multi-level cross-attentional masked autoencoder," 2022, *arXiv:2203.11725*.
[51] Y. Luo, Z. Chen, S. Zhou, and X. Gao, "Self-distillation augmented masked autoencoders for histopathological image classification," 2022, *arXiv:2203.16983*.
[52] J. Jiang, X. Lu, L. Zhao, R. Dazeley, and M. Wang, "Masked autoencoders in 3D point cloud representation learning," *IEEE Trans. Multimedia*, vol. 27, pp. 820–831, 2025.
[53] W. Kong, B. Liu, X. Bi, J. Pei, and Z. Chen, "Instructional mask autoencoder: A scalable learner for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1348–1362, 2024.
[54] J. Qi et al., "Masked spatial–spectral autoencoders are excellent hyperspectral defenders," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3012–3026, Jan. 2024.
[55] M. Gutmann and A. Hyvarinen, "Noise-contrastive estimation: A new estimation principle for unnormalized statistical models," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, 2010, pp. 297–304.
[56] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 815–823.
[57] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
[58] Y. Cai et al., "Large-scale hyperspectral image clustering using contrastive learning," 2021, *arXiv:2111.07945*.
[59] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
[60] F. Wang and H. Liu, "Understanding the behaviour of contrastive loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2495–2504.
[61] H. W. Kuhn, "The Hungarian method for the assignment problem," *Nav. Res. Logistics (NRL)*, vol. 52, no. 1, pp. 7–21, Feb. 2005.
[62] G. Jiang, Y. Zhang, X. Wang, X. Jiang, and L. Zhang, "Structured anchor learning for large-scale hyperspectral image projected clustering," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 3, pp. 2328–2340, Mar. 2025.
[63] L. van der Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008.



Mengjiao Zhang received the B.S. degree from the School of Computer Science, Liaocheng University, Liaocheng, China, in 2021. She is currently pursuing the M.S. degree with the University of Jinan (UJN), Jinan, China.

Her research interests include multiview learning and subspace clustering.



Tianhao Han received the M.S. degree from Shandong Provincial Key Laboratory of Network Based Intelligent Computing, University of Jinan, Jinan, China, in 2021. He is currently pursuing the Ph.D. degree with the Department of Computer Science and Engineering, MoE Laboratory of AI, Shanghai Jiao Tong University (SJTU), Shanghai, China.

His research interests include deep learning, remote sensing image processing, hand pose estimation, and 3-D reconstruction.



Xiaofeng Qu received the B.Eng. degree from the University of Jinan, Jinan, China, in 2019, and the Ph.D. degree from Shandong Normal University, Jinan, in 2024.

From 2020 to 2024, she was with Shandong Normal University under the supervision of Prof. Huaxiang Zhang. She is currently a Lecturer with the School of Information Science and Engineering, University of Jinan. Her research interests include machine learning, pattern recognition, and information retrieval.



Xizhan Gao received the Ph.D. degree in control science and engineering from Nanjing University of Science and Technology, Nanjing, China, in 2019.

He is currently a Lecturer with the School of Information Science and Engineering, University of Jinan, Jinan, China. His research interests include pattern recognition, computer vision, and video processing.



Xinwang Liu (Senior Member, IEEE) received the Ph.D. degree from the National University of Defense Technology (NUDT), Changsha, China, in 2013.

He is currently a Professor with the School of Computer, NUDT. He has published more than 100 peer-reviewed papers, including those in highly regarded journals and conferences, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, ICML, NeurIPS, ICCV, CVPR, AAAI, and IJCAI. His research interests include kernel learning and unsupervised feature learning.

Dr. Liu serves as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS and *Information Fusion Journal*. More information can be found at https://xinwangliu.github.io/



Sijie Niu (Member, IEEE) received the B.S. degree from the School of Computer Science, Liaocheng University, Liaocheng, China, in 2007, and the Ph.D. degree from Nanjing University of Science and Technology, Nanjing, China, in 2016.

He was a Visiting Scholar with Stanford University, Stanford, CA, USA, in 2014. He currently holds a post-doctoral position in medical image analysis at the University of North Carolina System (UNC), Chapel Hill, NC, USA. He is currently an Associate Professor with the School of Information Science and Engineering, University of Jinan, Jinan, China. His research interests include pattern recognition, machine learning, image processing, and medical image analyses.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:12:21 UTC from IEEE Xplore. Restrictions apply.